Dietary Choline Intake Adequacy and Liver Enzymes by BMI in U.S. Adults: A Cross-Sectional Analysis of NHANES 2021-2023

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To my wife, Alexis, thank you for believing in me even when I didn't believe in myself. To our Penny, thank you for being the best dog in the world and helping me take breaks from the computer screen.

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Ethical approval and informed consent statements

NHANES protocol was approved by the National Center for Health Statistics (NCHS) Ethics Review Board. Secondary analysis of publicly available, de-identified data did not require institutional review board approval. All NHANES participants provided written informed consent.

Consent for publication

Not applicable

Data availability statement

All code, extracted data, and supplementary analyses can be accessed at https://github.com/LouieNeri/choline-bmi-nhanes

Authors' contributions

ALN: Conceptualization, Methodology, Formal analysis, Software, Visualization, Writing - original draft. TG: Supervision, Methodology, Writing - review and editing. Both authors read and approved the final manuscript.

Abstract

Background: Adequate Intake (AI) of choline was established in 1998 through clinical trials on healthy adults. Since then, the prevalence of obesity has skyrocketed, and it remains unknown whether the consumption of choline, or the relationship between choline and liver health, varies according to BMI.

Aim: To estimate the prevalence of inadequate choline intake according to BMI categories, and to examine whether the co-occurrence of low choline intake and obesity is related to liver enzymes.

Methods: In a sample of 4,124 adults from NHANES, August 2021 to August 2023, with at least two valid 24-h dietary recalls, survey-weighted proportions of adults consuming below the AI were calculated for each BMI category. In a sample of 2,043 adults with no liver disease, hepatitis B, or excessive alcohol consumption, survey- weighted regressions examined ALT, AST, and GGT in adults with both obesity and low choline intake vs all other adults.

Results: Overall, 84.4% (95% CI 82.7-86.0) consumed below the AI, without differences across BMI categories (P = 0.81). There were no differences in ALT, AST, and GGT of adults with obesity and low choline intake vs other adults after multiple testing adjustments.

Conclusion: Most U.S. adults do not meet the AI of choline, regardless of BMI. Liver enzymes were unable to differentiate between low and high levels of choline intake, even among people with obesity.

Keywords: choline, Adequate Intake, obesity, body mass index, liver enzymes, dietary reference intakes, NHANES

Introduction

The Adequate Intake (AI) for choline was derived from a small number of controlled depletion-repletion studies in otherwise healthy U.S. adults. In these studies, the removal of dietary choline led to hepatic steatosis and elevated serum liver enzymes that resolved with the reintroduction of dietary choline (Fischer et al., 2007; Institute of Medicine, 1998; Zeisel et al., 1991). Because these data were limited, the Institute of Medicine could not establish an Estimated Average Requirement (EAR), and thus the AI was established in 1998 to be 550 mg/day for men and 425 mg/day for women, which has not been updated since then.

National data indicate that the majority of U.S. adults do not meet the AI for choline (Freedman et al., 2024; Wallace et al., 2018). Simultaneously, the prevalence of obesity among U.S. adults has nearly doubled since the era in which the AI was established (DeCleene et al., 2026) This population-wide change also infers the potential that the AI for choline still applies to adults with higher BMIs rather than the healthy-weight adults in whom it was originally derived.

The liver synthesizes phosphatidylcholine via the phosphatidylethanolamine N-methyltransferase (PEMT), supplying approximately one-third of hepatic requirements, providing a buffer to lower dietary intake (Vance, 2013). However, this endogenous buffer mechanism may be compromised in obesity. Individuals with severe obesity have diminished liver PEMT activity (Kartsoli et al., 2020), and adults with obesity have lower serum choline than their healthy-weight counterparts (Winslow et al., 2025). It is possible that obese individuals depend more on dietary intake to meet their choline needs, and hence impaired liver function from inadequate intake may be more severe in this population. Obesity and choline needs in obesity as well as choline intake in free-living populations have been identified as research priorities at the 2025 Choline Research Symposium (Paules et al., 2025).

Adults with obesity may be at increased risk for inadequate choline intake possibly due to dietary patterns. Namely, if inadequate choline intake is related to liver injury biomarkers in adults with obesity, the current AI may not be adequate to protect against liver injury. Although a study using the National Health and Nutrition Examination Survey (NHANES) data choline intake and liver enzymes levels, with body mass index (BMI) adjusted as a covariate, were reported with BMI not as a focus for the examination (Wallace et al., 2025). No study has investigated either of these possibilities in a nationally representative sample.

Thus, there were two aims of this study: 1) estimate the prevalence of inadequate choline intake among adults, stratified by BMI, sex and, among women, menopausal status; and 2) determine the relationships of inadequate choline intake and obesity with serum liver enzymes: alanine amino transferase (ALT), aspartate aminotransferase (AST), and gamma-glutamyl transferase (GGT).

Methods

We conducted a cross-sectional analysis of data from the National Health and Nutrition Examination Survey (NHANES) 2021 to 2023 cycle. NHANES is conducted by the National Center for Health Statistics (NCHS) and is a complex, stratified, multistage probability sample of the civilian, non-institutionalized, U.S. population (National Center for Health Statistics, 2024a). NHANES participants complete a household survey as well as a physical exam, which includes body measurements and a blood draw.

The survey was approved by the NCHS Ethics Review Board, and all participants provided written informed consent. This secondary analysis of publicly available, deidentified data was not subject to ethics review. Reporting was conducted in accordance with the

Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for cross-sectional studies (von Elm et al., 2008).

Dietary intake was assessed by two 24-hour dietary recalls using the USDA Automated Multiple-Pass Method (Moshfegh et al., 2008). Choline and total energy (kcal/d) values were derived using the USDA Food and Nutrient Database for Dietary Studies (US Department of Agriculture, Agricultural Research Service, n.d.). Only reliable recalls from NCHS, meeting the minimum data quality standards on both days, were included in the analyses and intake was calculated as the average of the two recall days.

For aim 1, mean intake was compared to the sex-specific AI of 550 mg/d for men and 425 mg/d for women (Institute of Medicine, 1998). We classified participants as either below the AI or above the AI.

For aim 2, choline was expressed as density (mg per 1,000 kcal) and divided into quartiles within the aim 2 sample. Density was used because absolute choline intake largely tracks total food intake. Choline intake was modeled as the density of choline intake (mg/1,000 kcal) and also divided into four quartiles in the aim 2 sample. Choline density was used because choline intake is strongly correlated with other measures of food intake, including total energy intake ($r = 0.66$), in this population. In contrast, choline density was not correlated with energy intake ($r = -0.13$). Choline density accounts for the dietary concentration of choline and removes the influence of total food intake. In the aim 2 sample, quartile cutoffs were 121.52, 156.53, 202.23 mg/1,000 kcal.

BMI was calculated from measured height and weight and grouped as normal (18.5 to 24.9 kg/m^2), overweight (25.0 to 29.9 kg/m^2), or obese (30.0 kg/m^2 or greater). Underweight

adults (BMI below 18.5 kg/m²) were excluded from BMI comparisons as the group was too small for stable weighted estimates.

The outcome in aim 1 was inadequate choline intake (dietary intake less than AI). The outcomes for aim 2 were serum concentrations of the liver enzymes alanine aminotransferase (ALT), aspartate aminotransferase (AST), and gamma glutamyltransferase (GGT) in the standard biochemistry panel. The values were log-transformed to reduce the skewness of the distribution to the right.

In adult women, menopausal status was derived from questions in the reproductive health questionnaire (National Center for Health Statistics, 2024c). Women who reported not having menstrual periods for 1 year or longer due to menopause were coded as postmenopausal, and those who still menstruated regularly were coded as premenopausal. Those who did not have periods because of hysterectomy were assigned no status ("not applicable") as the public use file for this cycle does not include information to determine whether ovaries were present.

The aim 1 sample of adults aged ≥ 20 years included those who provided two good-quality 24-hour dietary recalls, had a valid 2-day food weight record, and had a measured BMI. Adults with implausible estimated energy intake were excluded from analyses: those with a mean of intake <500 kcal/day or >8,000 kcal/day for men, and <500 kcal/day or >5,500 kcal/day for women. The values above are wider than those often cited, which range between 500 to 3,500 kcal/day for females and 800 to 4,200 kcal/day for males, since individuals with obesity can require energy intakes that go beyond the upper limit (Banna et al., 2017).

The aim 2 sample included those with a measured ALT value and those without self-reported liver disease, hepatitis B infection, and heavy alcohol use, defined as >2 drinks/day for men and >1 drink/day for women. Heavy alcohol use and liver disease are known to elevate liver

157 enzymes independent of choline intake. Reported hepatitis C infection was not included in the
158 public use file for this cycle.

159 In both aims, the two-day dietary sample weights were used together with survey stratum
160 and primary unit identifiers, following NCHS analytic guidelines (National Center for Health
161 Statistics, 2024b). The survey design was defined using the full dataset, and subgroups were
162 specified by subsetting the survey design, which maintained the full variance estimation.
163 Standard errors were computed using Taylor series linearization (Lumley, 2004).

164 For aim 1, the proportion of adults with inadequate choline intake (<AI) was estimated
165 using the two-day dietary sample weights for the full sample and stratified by BMI category, sex
166 and BMI category, and for women by menopausal status and BMI category. Rao-Scott chi-
167 square tests were used to determine if proportions varied across BMI categories.

168 For aim 2, there were only 15 degrees of freedom in this cycle, which limits the ability to
169 estimate complex models, as testing for the main effects of choline quartile and BMI and their
170 interaction would overfit the data and was therefore not pursued. Instead, a single a priori
171 contrast was defined between adults in the lowest quartile of choline density and obese adults
172 with all other subjects to evaluate the hypothesis that dietary inadequacy is most important when
173 endogenous production is also low. Models were fit using the two-day dietary sample weights
174 for each of the log-transformed liver enzymes using linear regression, adjusting for age, sex,
175 race/ethnicity, and family income-to-poverty ratio. Regression coefficients were transformed to
176 ratios of geometric means and 95% confidence intervals. Bonferroni adjustment was used to
177 evaluate statistical significance, resulting in a required level of significance of $P < 0.017$.

178 A secondary analysis was performed with BMI categories (normal, overweight, and
179 obese) and a dichotomous variable for low choline intake, where the reference group was normal

BMI adults with high choline intake. Six groups were formed, where each was a combination of BMI category and low-choline intake status. Joint Wald tests were performed to evaluate differences in enzymes across the six groups. A second model was fitted with BMI category, low choline intake status, and their interaction to evaluate differences between low and high choline intake for each BMI category.

The code and data for the analyses are available at https://github.com/LouieNeri/choline-bmi-nhanes. Analyses were performed using R version 4.3.3 (R Core Team, 2024) with the following packages: nhanesA (Ale et al., 2024), survey (Lumley, 2004), tidyverse (Wickham et al., 2019), gt (Iannone et al., 2024), gtsummary (Sjoberg et al., 2021), and ggthemes (Arnold, 2024).

Results

A total of 7,809 participants in NHANES between August 2021 and August 2023 were adults aged 20 years and older. In aim 1, 4,124 adults were included in analyses after excluding 3,524 participants without two reliable dietary recalls, 47 without a measured BMI, 24 with implausible energy intake, 36 who were pregnant or breastfeeding, and 54 who were underweight. In aim 2, an additional 371 participants without ALT values were excluded, as well as 228 with liver disease, 37 with hepatitis B, 1,180 with heavy alcohol consumption, and 265 with missing covariate data. The final sample size for aim 2 was 2,043 adults

Participants in Aim 1 included 4,124 adults (2,288 women and 1,836 men) with a mean (SD) age of 49.2 (17.1) years and a BMI distribution of 24.4% normal weight, 33.5% overweight, and 42.0% obesity. The mean daily choline intake was 334.7 mg for all adults, 384.5 mg for men, and 285.6 mg for women (Table 1).

For all adults combined, 84.4% (95% CI: 82.7-86.0%) were below the AI for choline and this proportion did not differ by BMI (P = 0.81), with 83.5% (95% CI: 80.2-86.7%), 84.9% (95% CI: 82.4-87.5%), and 84.4% (95% CI: 81.1-87.8%) of normal weight, overweight, and obese adults below the AI, respectively (Table 2). Among women, 81.4% (95% CI: 78.4-84.4%), 86.1% (95% CI: 81.9-90.2%), and 87.5% (95% CI: 83.3-91.8%) were below the AI for normal weight, overweight, and obesity, respectively (P = 0.16). Among men, the prevalence of choline intake below the AI did not differ by BMI (P = 0.24), with 85.8%, 84.0%, and 81.1% below the AI in the normal weight, overweight, and obesity groups, respectively. The prevalence of choline intake below the AI was higher at higher BMI among premenopausal women (78.4% [95% CI: 73.7-83.1%], 82.5% [95% CI: 75.1-89.9%], and 90.4% [95% CI: 85.4-95.4%] of normal weight, overweight, and obese premenopausal women had choline intakes below the AI). Among postmenopausal women, there was no clear pattern, with 83.2%, 89.0%, and 82.2% below the AI at normal weight, overweight, and obesity, respectively. The BMI pattern did not differ significantly by menopausal status (P for interaction = 0.076; Table 3).

Aim 2 analysis sample consisted of 2,308 adults where 2,043 had covariate information and formed the group for adjusted analyses (2,039 for AST). The ratio of geometric means (GMR) of adults with obesity and low choline intake compared to all others was 1.04 (95% CI: 0.97-1.12; P = 0.24) for ALT, 0.97 (95% CI: 0.90-1.03; P = 0.28) for AST, and 1.17 (95% CI: 1.03-1.33; P = 0.018) for GGT (Table 5). None of these met the corrected significance level of P < 0.017.

In the secondary analysis, the sample size in the six groups ranged from 136 to 680 adults. ALT and GGT differed across the six groups (P = 0.002 for each), but AST did not (P = 0.14), with estimates following BMI category (Table 6, Figure 3). There were no differences in

liver enzymes between adults with low vs higher choline intake within the BMI category among normal weight or overweight adults. However, among obese adults, those with low choline intake had lower ALT (GMR 0.88; 95% CI: 0.81-0.96; P = 0.011), but not AST (GMR 0.94; 95% CI: 0.87-1.01; P = 0.062) or GGT (GMR 0.95; 95% CI: 0.83-1.09; P = 0.36) compared to those with higher choline intake (Table 7).

Discussion

In support of previous evidence, 84.1% of adults consuming less than the AI for choline was slightly lower than the 90% generally cited, and similar to that reported in other analyses of NHANES data (Freedman et al., 2024; Wallace et al., 2018). However, contrary to our initial hypothesis, we did not find evidence of substantially higher liver enzymes (ALT, AST, and GGT) in obese adults with low choline intake as compared to other adults. Only GGT was higher in the obesity/low choline group, but this difference was not statistically significant after correction for multiple testing. The lower prevalence is likely due to the method of dietary assessment. The 2-day recall used in our analysis retains more of the day-to-day variability in food intake than other methods, expanding the distribution and including more extreme values than usual intake methods (Dodd et al., 2006). This would affect all BMI categories in the same way, so it does not explain the lack of difference between the BMI categories.

The main finding for aim 1 was that the gradient in choline intake across BMI categories was near zero. If obese adults had consumed less choline, then the lower circulating choline in these individuals would be due to dietary intake. However, they did not consume less choline than other adults, and this was true for both absolute and density measures of choline intake. The lower circulating choline levels of obese adults is, therefore, unlikely to be due to differences in

choline intake. Other mechanisms, such as differences in absorption, liver metabolism or internal production of choline, might be responsible. For example, lower expression of PEMT in the liver of adults with severe obesity (Kartsoli et al., 2020), which could not be assessed in this study, might be one explanation. We did find a trend of a higher estimated prevalence of insufficient choline intake by BMI in premenopausal women. Since estrogen stimulates PEMT production (Fischer et al., 2007) premenopausal women may be able to synthesize more choline endogenously than postmenopausal women, which may increase the risk However, more direct research is needed to confirm this potential mechanism.

Our null finding for aim 2 is consistent with a recent analysis of NHANES data that found low choline intake was not associated with liver enzymes in the U.S. population in general (Wallace et al., 2025). We extended that finding by examining whether obesity, which may lower endogenous choline synthesis, would change the nature of the association.

There are two possible explanations for our aim 2 finding. First, liver enzymes, the serum aminotransferases, may be too insensitive to detect differences in chronic choline intake. For example, in a controlled feeding trial, plasma choline and betaine levels distinguished between low intake and intakes at the AI, but serum liver enzymes did not (Trujillo-Gonzalez et al., 2026). Second, studies that have used a direct measure of liver fat also found no association between choline intake and liver fat. For example, a cross-sectional analysis of NHANES data from 2017-2020, using FibroScan to measure liver fat, found no association between total choline intake and presence of metabolic dysfunction-associated steatotic liver disease (Taesuwan et al., 2025).

Our secondary analysis also showed a statistically significant difference in liver enzymes for ALT and GGT. The estimates followed the pattern for BMI category, not choline intake, which is expected based on the known associations between fat and liver enzymes.

This study had several strengths. We used the latest NHANES data and included both measured BMI and liver enzymes. We used sample weights to provide a nationally representative sample of U.S. adults. The study design was in accord with the methodology guidelines provided by the NCHS for the full sample. We chose a single contrast that was pre-specified to ensure we stayed within the available degrees of freedom.

However, there are several limitations. A cross-sectional study design does not allow for inferences of causality. The dietary intake values reported were averaged between the 2-day recall rather than being modeled as a distribution of usual intake, which limits the ability to estimate true prevalence rates for individuals across the entire range of choline intake. Individuals with higher BMI are known to underreport their intakes more frequently, which could attenuate the difference in reported intakes between the groups (Moshfegh et al., 2008). In this study, there was a lack of degrees of freedom to allow for testing an interaction model of choline intake and alcohol intake, whereas the degrees of freedom for all 6 groups was very small with df =5 for the denominator. Heavy alcohol intake was classified based on the average number of drinks per drinking day, which may have excluded some individuals who drank moderate volumes of alcohol per day but drank less frequently, which decreased the sample size tested in aim 2. Additional study limitations were related to data availability, such as the inability to exclude Hepatitis C infection, the lack of an ideal measure of body fat for BMI, the lack of direct liver fat measurements, and the lack of a menopausal status that could be assigned for women who reported a hysterectomy.

Conclusions

It appears that dietary choline intake levels are not different across BMI categories, and liver enzymes available in national population surveys do not appear to reflect habitual intake. A controlled feeding study across a broad range of BMIs is needed to determine whether there are differences in choline requirements in obesity, with circulating choline and choline metabolites or direct measures of liver fat content as outcomes.

Tables and Figures

Table 1. Baseline characteristics by sex

Characteristic	Sex			
	Female (n = 2,288)[1]	Male (n = 1,836)[1]	Total (n = 4,124)[1]	P value[2]
Age, years	50.1 (17.2)	48.3 (17.0)	49.2 (17.1)	0.033
Race and ethnicity				0.5
Mexican American	139 (6.6%)	112 (7.6%)	251 (7.1%)	
Other Hispanic	220 (9.0%)	170 (9.9%)	390 (9.4%)	
Non-Hispanic White	1,435 (61.6%)	1,160 (62.6%)	2,595 (62.1%)	
Non-Hispanic Black	267 (12.4%)	196 (9.8%)	463 (11.1%)	
Non-Hispanic Asian	95 (5.4%)	78 (5.2%)	173 (5.3%)	
Other Race - Including Multi-Racial	132 (5.0%)	120 (5.0%)	252 (5.0%)	
Education				0.10
Less than 9th grade	83 (2.7%)	73 (3.1%)	156 (2.9%)	
9-11th grade (Includes 12th grade with no diploma)	139 (5.5%)	141 (6.4%)	280 (5.9%)	
High school graduate/GED or equivalent	415 (21.7%)	388 (25.6%)	803 (23.6%)	
Some college or AA degree	752 (31.0%)	509 (29.0%)	1,261 (30.0%)	
College graduate or above	899 (39.1%)	725 (35.9%)	1,624 (37.5%)	
Income-to-poverty ratio	3.0 (1.7)	3.3 (1.6)	3.1 (1.6)	0.010
BMI category				0.094
Normal	578 (25.8%)	430 (23.1%)	1,008 (24.4%)	
Overweight	678 (31.1%)	668 (36.0%)	1,346 (33.5%)	
Obese	1,032 (43.1%)	738 (40.9%)	1,770 (42.0%)	
Energy intake, kcal/day	1,776.8 (573.5)	2,337.7 (840.5)	2,055.3 (771.3)	<0.001
Choline intake, mg/day	285.6 (132.5)	384.5 (184.0)	334.7 (167.6)	<0.001
Below choline AI	1,948 (85.5%)	1,539 (83.2%)	3,487 (84.4%)	0.062

[1]Unweighted n (survey-weighted %) or survey-weighted mean (SD).

[2]t-test adapted to complex survey samples; chi-squared test with Rao & Scott's second-order correction

AI, Adequate Intake (550 mg/day for men, 425 mg/day for women); BMI, body mass index.

303 **Table 2. Percentage below the choline AI by sex and BMI**

	Female (n = 2,288)		Male (n = 1,836)		Total (n = 4,124)	
BMI category (kg/m²)	n/N[1]	% (95% CI)[2]	n/N[1]	% (95% CI)[2]	n/N[1]	% (95% CI)[2]
Normal weight (18.5–24.9)	478/578	81.4 (78.4, 84.4)	359/430	85.8 (81.0, 90.6)	837/1,008	83.5 (80.2, 86.7)
Overweight (25.0–29.9)	573/678	86.1 (81.9, 90.2)	558/668	84.0 (81.2, 86.8)	1,131/1,346	84.9 (82.4, 87.5)
Obesity (≥30.0)	897/1,032	87.5 (83.3, 91.8)	622/738	81.1 (78.3, 83.9)	1,519/1,770	84.4 (81.1, 87.8)
All BMI categories	1,948/2,288	85.5 (83.3, 87.7)	1,539/1,836	83.2 (81.4, 85.0)	3,487/4,124	84.4 (82.7, 86.0)
P value[3]		0.16		0.24		0.81

[1]Unweighted number below the AI / number of participants.

[2]Survey-weighted percentage.

[3]Rao-Scott chi-square test across BMI categories.

AI, Adequate intake (550 mg/day for men, 425 mg/day for women); BMI, body mass index; CI, confidence interval.

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305 **Table 3. Percentage below the choline AI by menopausal status and BMI**

	Postmenopausal (n = 930)		Premenopausal (n = 649)	
BMI category (kg/m²)	n/N[1]	% (95% CI)[2]	n/N[1]	% (95% CI)[2]
Normal weight (18.5–24.9)	201/242	83.2 (79.6, 86.9)	163/208	78.4 (73.7, 83.1)
Overweight (25.0–29.9)	251/295	89.0 (84.8, 93.2)	151/185	82.5 (75.1, 89.9)
Obesity (≥30.0)	330/393	82.2 (74.4, 90.1)	223/256	90.4 (85.4, 95.4)
All BMI categories	782/930	84.8 (81.3, 88.3)	537/649	84.3 (81.0, 87.7)
P for interaction[3]		0.076		

[1]Unweighted number below the AI / number of participants.

[2]Survey-weighted percentage.

[3]BMI × menopausal status interaction from survey-weighted logistic regression. Women with a hysterectomy or unknown status were excluded.

AI, Adequate intake (425 mg/day for women); BMI, body mass index; CI, confidence interval.

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307 **Table 4. Liver enzymes by BMI category and choline quartile**

	Choline quartile (mg/1,000 kcal)[1]			
BMI category (kg/m²)	Q1 (<121.4)	Q2 (121.4–156.3)	Q3 (156.3–202.2)	Q4 (>202.2)
ALT, U/L				
Normal weight (18.5–24.9)	15.9 (14.7, 17.3)	17.0 (15.4, 18.7)	16.3 (15.3, 17.3)	17.3 (15.3, 19.6)
Overweight (25.0–29.9)	18.8 (16.9, 20.8)	17.8 (16.7, 19.0)	18.6 (17.2, 20.1)	19.0 (18.1, 20.0)
Obesity (≥30.0)	18.8 (17.7, 20.0)	22.4 (21.0, 24.0)	23.0 (21.3, 24.8)	20.7 (19.0, 22.6)
AST, U/L				
Normal weight (18.5–24.9)	20.0 (19.0, 21.1)	21.5 (20.4, 22.7)	20.1 (19.2, 21.1)	21.2 (19.7, 22.8)
Overweight (25.0–29.9)	19.8 (18.7, 21.1)	20.3 (19.2, 21.5)	20.1 (19.1, 21.2)	20.2 (19.2, 21.1)
Obesity (≥30.0)	19.6 (18.6, 20.7)	20.8 (20.0, 21.6)	21.5 (20.4, 22.6)	20.9 (20.3, 21.6)
GGT, U/L				
Normal weight (18.5–24.9)	16.5 (13.7, 19.9)	17.4 (15.2, 19.9)	17.5 (15.7, 19.5)	17.5 (15.1, 20.2)
Overweight (25.0–29.9)	19.6 (16.9, 22.8)	19.5 (17.0, 22.3)	21.5 (19.0, 24.2)	20.9 (18.8, 23.3)
Obesity (≥30.0)	23.2 (21.3, 25.3)	26.8 (24.1, 29.8)	24.7 (22.6, 27.1)	24.7 (22.7, 26.9)
Participants, n				
Normal weight (18.5–24.9)	136	161	158	137
Overweight (25.0–29.9)	188	200	192	203
Obesity (≥30.0)	253	216	227	237

[1]Survey-weighted geometric mean (95% CI).

ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; CI, confidence interval; GGT, gamma-glutamyl transferase.

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309 **Table 5. Low choline and obesity vs. all others**

Characteristic	ALT (n = 2,043)			AST (n = 2,039)			GGT (n = 2,043)		
	Ratio[1]	**95% CI[2]**	P value	Ratio[1]	**95% CI[2]**	P value	Ratio[1]	**95% CI[2]**	P value
Low choline and obesity (vs. all others)	1.04	0.97, 1.12	0.24	0.97	0.90, 1.03	0.28	1.17	1.03, 1.33	0.018

[1]Ratio of geometric means, adjusted for age, sex, race and ethnicity, and income-to-poverty ratio.

[2]CI = Confidence Interval

Low choline = lowest quartile of choline density.

ALT, alanine aminotransferase; AST, aspartate aminotransferase; CI, confidence interval; GGT, gamma-glutamyl transferase.

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311 **Table 6. Liver enzymes across BMI and choline groups**

Characteristic	ALT (n = 2,043)			AST (n = 2,039)			GGT (n = 2,043)		
	Ratio[1]	**95% CI**[2]	P value[3]	Ratio[1]	**95% CI**[2]	P value[3]	Ratio[1]	**95% CI**[2]	P value[3]
BMI category and choline intake									
Normal, higher choline	—	—		—	—		—	—	
Normal, low choline	0.95	0.83, 1.10	0.44	0.96	0.90, 1.03	0.18	0.94	0.76, 1.16	0.49
Overweight, higher choline	1.09	1.00, 1.19	0.049	0.96	0.91, 1.02	0.17	1.19	1.05, 1.34	0.017
Overweight, low choline	1.11	0.92, 1.33	0.22	0.95	0.85, 1.05	0.24	1.13	0.88, 1.45	0.26
Obese, higher choline	1.35	1.22, 1.48	<0.001	1.02	0.96, 1.08	0.41	1.50	1.35, 1.66	<0.001
Obese, low choline	1.19	1.07, 1.31	0.008	0.95	0.87, 1.04	0.24	1.42	1.17, 1.72	0.005

[1]Ratio of geometric means, adjusted for age, sex, race and ethnicity, and income-to-poverty ratio.

[2]CI = Confidence Interval

[3]Joint test across the six groups: ALT P = 0.002; AST P = 0.14; GGT P = 0.002.

Low choline = lowest quartile of choline density; higher choline = quartiles 2 to 4.

ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; CI, confidence interval; GGT, gamma-glutamyl transferase.

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313 **Table 7. Low vs. higher choline within each BMI category**

	ALT			AST			GGT		
BMI category (kg/m²)	Ratio[1]	95% CI	P value	Ratio[1]	95% CI	P value	Ratio[1]	95% CI	P value
Normal weight (18.5–24.9)	0.95	0.83, 1.10	0.44	0.96	0.90, 1.03	0.18	0.94	0.76, 1.16	0.49
Overweight (25.0–29.9)	1.01	0.87, 1.17	0.83	0.98	0.92, 1.06	0.59	0.95	0.74, 1.23	0.65
Obesity (≥30.0)	0.88	0.81, 0.96	0.011	0.94	0.87, 1.01	0.062	0.95	0.83, 1.09	0.36

[1]Ratio of geometric means for low vs. higher choline within the same BMI category, adjusted for age, sex, race and ethnicity, and income-to-poverty ratio.

Low choline = lowest quartile of choline density; higher choline = quartiles 2 to 4.

ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; CI, confidence interval; GGT, gamma-glutamyl transferase.

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Figure 1. Participant flow, NHANES August 2021 to August 2023.

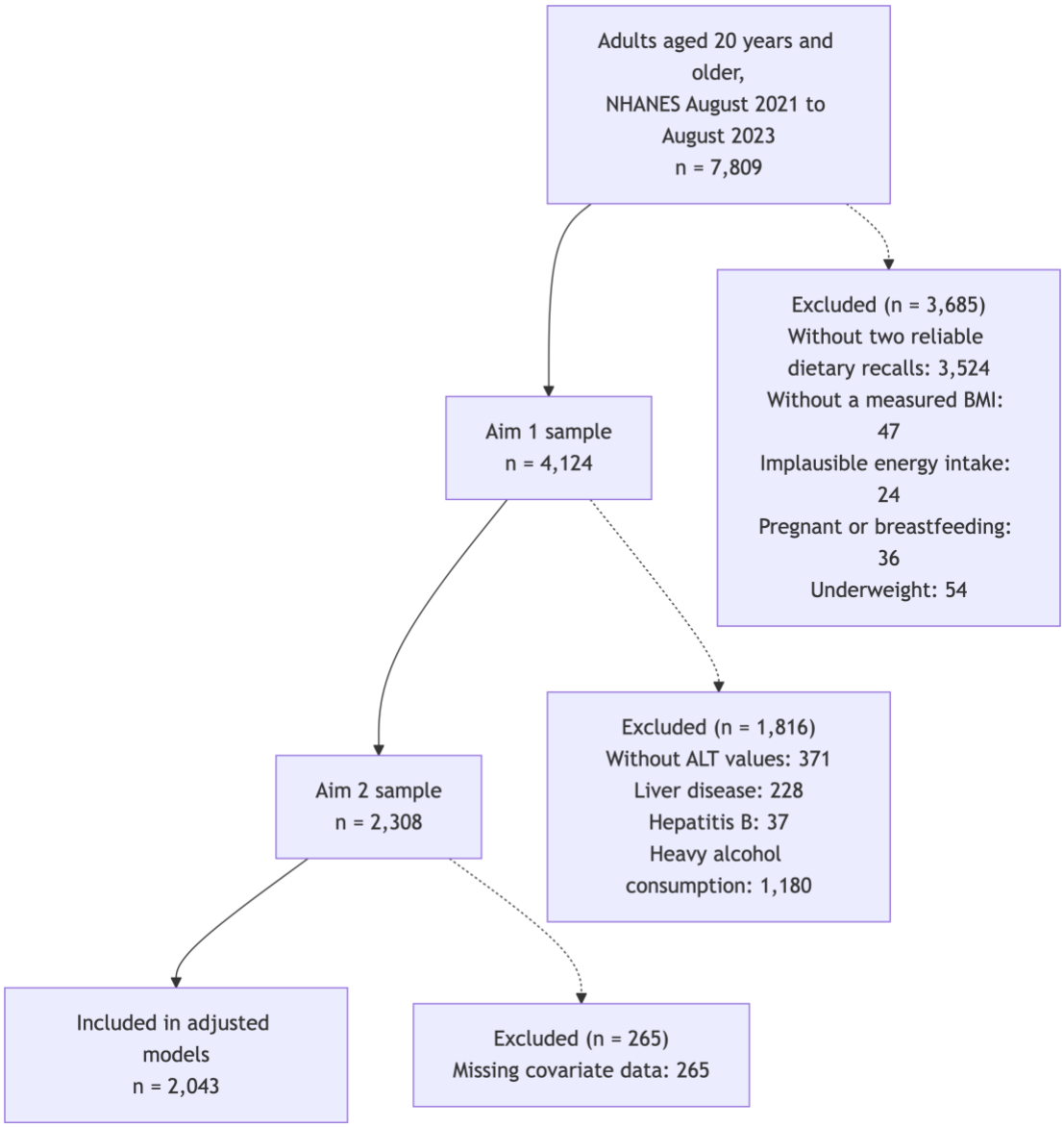


Figure 2

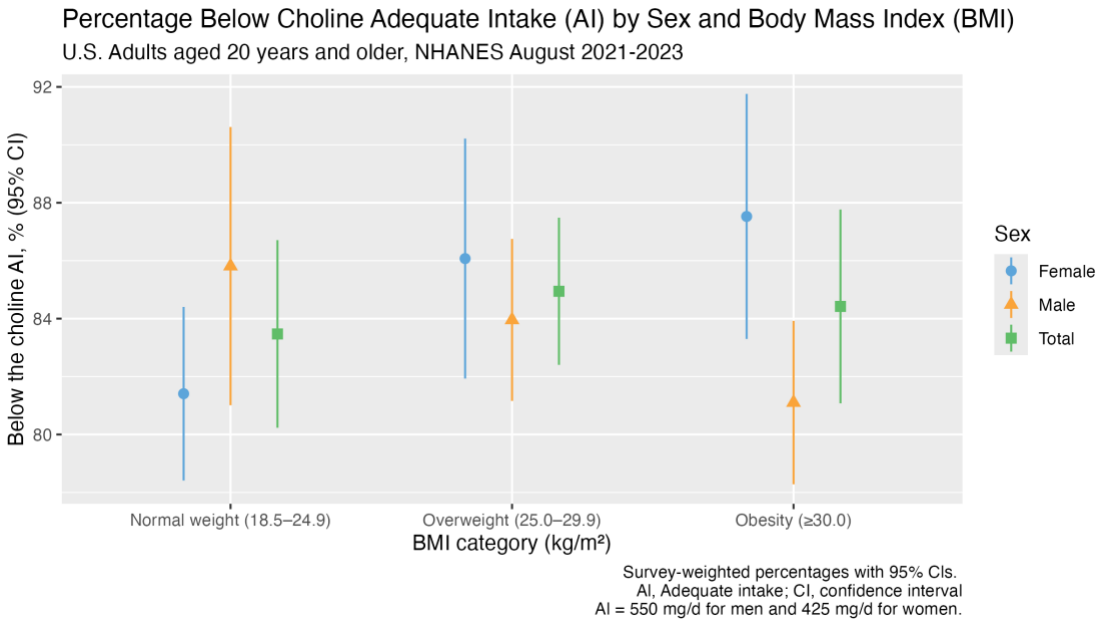

Percentage Below Choline Adequate Intake (AI) by Sex and Body Mass Index (BMI)
U.S. Adults aged 20 years and older, NHANES August 2021-2023
Below the choline AI, % (95% CI)
92
88
84
80
Normal weight (18.5–24.9)
Overweight (25.0–29.9)
Obesity (≥30.0)
BMI category (kg/m²)
Sex
Female
Male
Total
Survey-weighted percentages with 95% CIs.
AI, Adequate intake; CI, confidence interval
AI = 550 mg/d for men and 425 mg/d for women.

Figure 3

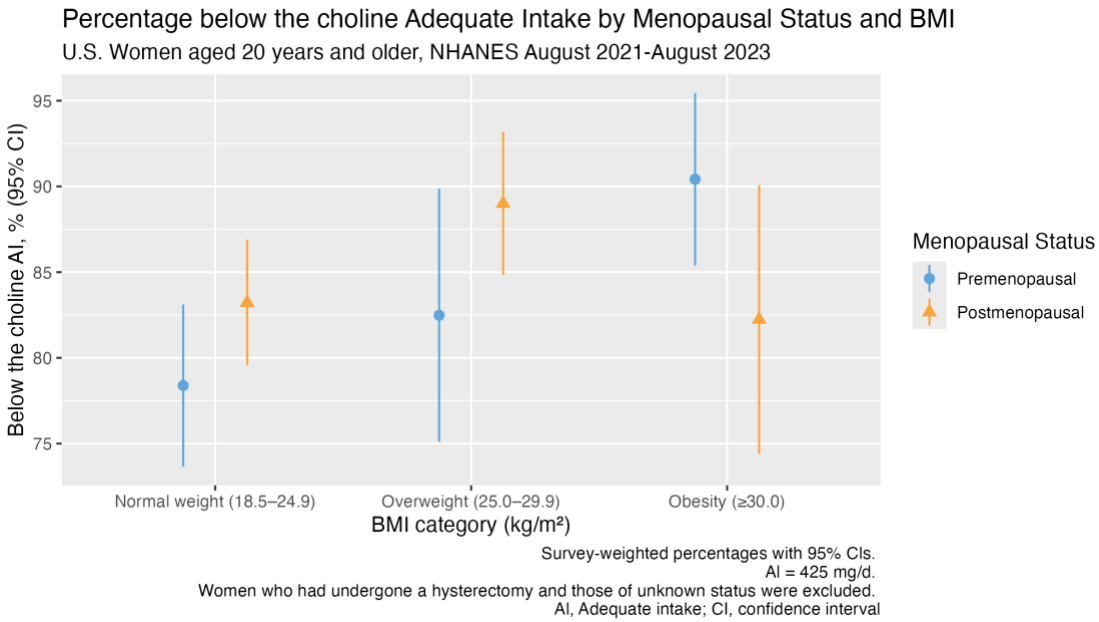
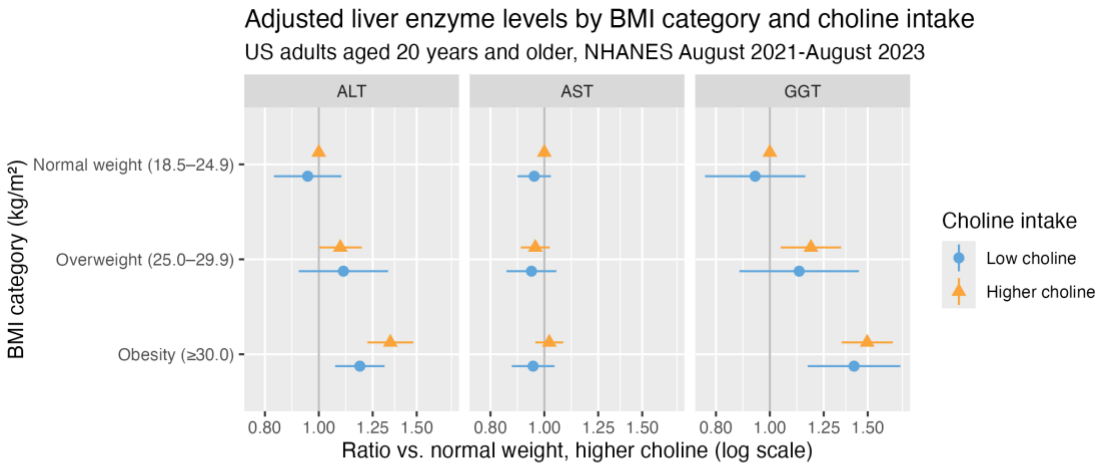

Percentage below the choline Adequate Intake by Menopausal Status and BMI
U.S. Women aged 20 years and older, NHANES August 2021-August 2023
Below the choline AI, % (95% CI)
95
90
85
80
75
Normal weight (18.5–24.9)
Overweight (25.0–29.9)
Obesity (≥30.0)
BMI category (kg/m²)
Menopausal Status
Premenopausal
Postmenopausal
Survey-weighted percentages with 95% CIs.
AI = 425 mg/d.
Women who had undergone a hysterectomy and those of unknown status were excluded.
AI, Adequate intake; CI, confidence interval

Figure 4



Ratios of geometric means with 95% CIs, adjusted for age, gender, race/ethnicity, and income-to-poverty ratio.
Low Choline: lowest quartile of choline density (<121.4 mg/1,000 kcal);
High Choline: quartiles 2 through 4.
ALT: alanine aminotransferase;
AST: aspartate aminotransferase;
BMI: body mass index;
CI: confidence interval;
GGT: gamma-glutamyl transferase

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